A-level

ENVIRONMENTAL SCIENCE

Paper 1 - Generated practice paper Set 33

Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

Marking guidance

• Each part opens with a line in bold saying how its marks are shared out, followed by one sentence on how to mark it.
• The figure in bold at the start of a marking point is the mark that point earns. Lines indented beneath it, starting with a dash, are the detail that belongs to the same mark; a line starting **OR** is another route to it.
• **Also accept** lists other wordings and answers that earn the mark - wording never has to match these notes, only the science. **Do not credit** lists the answers that look right and earn nothing.
• Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
• For levels-of-response questions, determine the level from the descriptors first, then position the mark within the level using the indicative content.
• Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	**Correct answer only** 1 mark for the correct option. No mark if more than one box is ticked. **1** C **Also accept** • **Critical Group** → the people most at risk / most heavily exposed	1
01.2	**1 mark each** 1 mark for each of the 2 points below. **1** Monitoring can be focused on the sites predicted to be most at risk / where the pollutant will concentrate **1** No need to sample sites that were never likely to be polluted – sampling also checks the accuracy of the predictions **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **accuracy** → close to the true value	2

Do not write outside the box

Question	Marking guidance	Marks
01.3	**1 mark each** 1 mark for each of the 2 points below. **1** Behaviour of new chemicals can be predicted from similar existing chemicals: it is likely to persist, bioaccumulate and biomagnify **1** So acceptable emission limits, control methods and monitoring programmes can be set before release (precautionary principle) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **persistent** → does not break down / stays in the environment for a long time **Do not credit** • the precautionary principle described as banning anything new - it shifts the burden of proof onto showing that harm is unlikely	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 243-245; AQA spec 3.4.3.1.	
02.1	**1 mark each** 1 mark for each of the 3 points below. **1** UV light splits diatomic oxygen (O2) into two monatomic oxygen atoms (O) **1** Monatomic oxygen reacts with diatomic oxygen to form ozone (O3) **1** UV light also splits ozone into O2 and O – the reactions form a dynamic equilibrium which absorbs UV (converting it to chemical energy / heat) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3
02.2	**Marks shared** Up to 6 marks in total - share them between the 4 points below according to how completely each one is made. **up to 2 Useful:** boiling points near ambient temperature so easily liquefied under low pressure (aerosol propellants, fridge / air-conditioning coolants) – dissolve grease but do not damage electrical components (solvents) – non-flammable – non-toxic **up to 2 Persistence:** chemically stable / unreactive in the troposphere so they remain long enough to be carried up to the stratosphere **up to 2 Dissociation by UV:** in the stratosphere higher UV levels break the carbon-chlorine bonds releasing chlorine free radicals **up to 2** Chlorine reacts with monatomic oxygen (and ozone), preventing ozone formation – chlorine is regenerated so one atom destroys many ozone molecules – bromine / iodine act similarly **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **CFCs** → CFCs / halogenated organic compounds releasing chlorine • **Persistence** → does not break down / stays in the environment for a long time **Do not credit** • answers that make ozone depletion a cause of global warming, or that describe CFCs as physically blowing a hole - chlorine atoms catalyse the breakdown of ozone and are regenerated	6

Do not write outside the box

Question	Marking guidance	Marks
02.3	**1 mark each** 1 mark for each of the 3 points below. **1** UVB energy is absorbed by cells and breaks up biological molecules / DNA causing mutations, skin cancer, cataracts, sunburn (human health) **1** Damage to leaf tissue and reduced photosynthesis / crop yields (plants) **1** Damage to marine algae, phytoplankton, corals and planktonic larvae (marine organisms), reducing productivity and food webs **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks.	3
02.4	1 mark for the point below. **1** An area where the total ozone column falls below 220 Dobson Units (normal is about 300 DU) - a level never recorded before CFC pollution **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	1
02.5	1 mark for the point below. **1** Concentration varies with altitude and the stratosphere is not normally sampled – Dobson Units measure the total column thickness instead **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • answers that make ozone depletion a cause of global warming, or that describe CFCs as physically blowing a hole - chlorine atoms catalyse the breakdown of ozone and are regenerated	1
02.6	1 mark for the point below. **1** The British Antarctic Survey at Halley detected higher UV at ground level, showing ozone above had decreased (satellites above the stratosphere cannot measure downward UV directly) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • answers that make ozone depletion a cause of global warming, or that describe CFCs as physically blowing a hole - chlorine atoms catalyse the breakdown of ozone and are regenerated	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 108-109; AQA spec 3.2.1.3.	

Do not write outside the box

Question	Marking guidance	Marks
03.1	**2 marks per point** 2 marks for each of the 3 points below; award 1 mark where a point is only partly made. **2** Aerodynamic shapes / fairings reduce friction with the air (air resistance) so less energy is needed to propel the vehicle, especially at speed – hydrodynamics for ships **2** Reduced mass (plastics, carbon-fibre composites, high-strength steel alloys, aluminium engine blocks, lighter batteries and magnets, rounded shapes) means less energy to accelerate and less rolling resistance, so lower fuel consumption **2** Conventional friction brakes convert kinetic energy to waste heat – regenerative braking uses a generator to convert it to electricity stored in a battery (or returned to the rail grid) which then propels the vehicle, so less fuel is needed to accelerate again **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	6
03.2	**1 mark each** 1 mark for each of the 2 points below. **1** Pneumatic tyres deform as they rotate – the continual change in shape and air movement inside generates frictional heat (more if under-inflated) **1** Steel wheels do not deform so rolling resistance is very low **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
03.3	**Any 2 of 4** 1 mark each for any 2 of the 4 points below, in any order, maximum 2. **1** More valves per cylinder to remove exhaust gases fully so they do not mix with fresh fuel **1** Better temperature control (thermostatic fan) **1** Electronic ignition control / regular servicing to keep spark timing correct **1** Turbochargers supplying more oxygen **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 225-227; AQA spec 3.3.4.2.1.	

Do not write outside the box

Question	Marking guidance	Marks
04.1	**1 mark each** 1 mark for each of the 2 points below. **1** Friction with the prevailing south-westerly winds drags surface water north-eastwards from the tropical Atlantic **1** Water in the NE Atlantic cools, becomes denser (and more saline as ice forms) and sinks, drawing warm surface water in to replace it – the cold water returns southwards at depth (thermohaline circulation) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **thermohaline** → the ocean conveyor driven by temperature and salinity **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks.	2
04.2	**1 mark each** 1 mark for each of the 4 points below. **1** Higher temperatures melt land ice on Greenland which flows into the North Atlantic **1** The fresh water dilutes the seawater, reducing its salinity and density **1** Less dense water sinks less easily so the flow rate of the current is reduced **1** Less heat is transported to NW Europe so the UK could become colder (like Moscow / Churchill at the same latitude) even while global temperatures rise **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	4
04.3	**1 mark each** 1 mark for each of the 3 points below. **1** Several interconnected systems act at once over different timescales: jet stream changes may raise temperatures, a slowing Conveyor may lower them, greenhouse gases raise them **1** Natural fluctuations (solar output, orbit) can hide or exaggerate anthropogenic change – individual events do not show trends **1** Time delays between cause and effect (oceans take a long time to warm because of volume and heat capacity) – incomplete understanding of the processes **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **anthropogenic** → human-caused / caused by human activity	3
04.4	1 mark for the point below. **1** Argo floats sink to a set depth for about 10 days, then surface and transmit temperature, salinity and position data – the sequence of positions shows current direction and speed (surface currents also by satellites and buoys) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 96-97; AQA spec 3.2.1.2, 3.2.2.3.	

Do not write outside the box

Question	Marking guidance	Marks
05.1	**1 mark each** 1 mark for each of the 2 points below. **1** Deaths rise as smoke concentration rises, peaking at about 930 per day on 8 Dec when smoke was highest (4.5 mg m^{-3}) **1** Deaths fall as smoke falls but remain above the starting level for several days (time lag / delayed effect) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
05.2	**Calculation** A correct final answer scores full marks on its own. If the answer is wrong, each correct step of working below earns its mark. **1** (930 - 250) / 250 x 100 **1** = 272% **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (%) is already printed on the answer line. **Do not credit** • A bare wrong answer with no working.	2
05.3	**1 mark each** 1 mark for each of the 3 points below. **1 Anticyclonic conditions:** clear skies allowed rapid radiation of heat from the ground at night so the ground and air near it cooled **1** Low wind velocity so air layers did not mix – London lies in a valley / basin where cold dense air collected: a temperature inversion formed **1** Smoke from coal fires was trapped below the inversion – moist air cooled to its dew point forming fog – smoke + fog = smog – the smog's high albedo prolonged the inversion **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **temperature inversion** → a layer of warm air above cooler air that traps pollutants • **albedo** → reflectivity / the proportion of solar radiation reflected / how much sunlight is bounced back **Do not credit** • 'ice reflects the heat back' - albedo is the reflection of incoming short-wave solar radiation, before it is absorbed	3

Do not write outside the box

Question	Marking guidance	Marks
05.4	**1 mark each** 1 mark for each of the 2 points below. **1** Smoke particles (PM10) are inhaled deeply and irritate the lungs / kill cilia so particles and bacteria are not cleared, causing bronchitis / infections **1** These groups have weaker lungs / immune systems / less reserve so the extra stress (with sulfur dioxide acting synergistically) was fatal **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **PM10** → fine solid particles / smoke and dust	2
05.5	1 mark for the point below. **1** Clean Air Act (1956) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 245-247; AQA spec 3.4.3.2.1.	
06.1	**1 mark each** 1 mark for each of the 3 points below. **1** Each regional population has its own unique gene pool / DNA profile **1** DNA from seized timber or ivory is compared with the database of regional populations **1** This identifies where the material came from, showing whether it was legally harvested and where poaching occurs **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **gene pool** → the range of alleles / genes present in a population	3
06.2	**1 mark each** 1 mark for each of the 2 points below. **1** Radio / GPS transmitters use radio waves - work in air and fresh water, giving positions (GPS stores or transmits location) **1** Radio waves do not travel through seawater so acoustic transmitters (sound) are used for marine animals, giving real-time positions **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • The converse of any point, ie the same comparison written the other way round.	2

Do not write outside the box

Question	Marking guidance	Marks
06.3	**Marks shared** Up to 3 marks in total - share them between the 2 points below according to how completely each one is made. **up to 2** Drones / ROVs, AUVs (eg Autosub), balloons, aircraft, satellites, animals (tags) **up to 2 Drones:** cheap, low disturbance, high-resolution images of areas hard to reach (cliffs, canopy, ice), repeated surveys – infrared cameras detect animals at night **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **infrared** → heat radiation / long-wave radiation re-emitted by the Earth **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks.	3
06.4	**1 mark each** 1 mark for each of the 2 points below. **1** Cameras record automatically (visible light by day, infrared at night) without researchers being present **1** So animals behave naturally and nests are not deserted – monitoring continues 24 hours **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **infrared** → heat radiation / long-wave radiation re-emitted by the Earth	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 404-406; AQA spec 3.7.2.3.	
07.1	**Correct answer only** 1 mark for the correct option. No mark if more than one box is ticked. **1** B - Tropospheric ozone (formed by reactions of NOx in sunlight)	1
07.2	**1 mark each** 1 mark for each of the 4 points below. **1 Primary pollutant:** released directly by human activities, eg sulfur dioxide from burning coal / smelting sulfide ores – NOx from high-temperature combustion **1 Secondary pollutant:** produced by chemical reactions between primary pollutants, often with non-pollutants (water, oxygen, sunlight) **1** Sulfur dioxide is oxidised by ozone to sulfur trioxide which dissolves to form sulfuric acid – NOx dissolve to form nitric acid **1** The acids (sulfurous, sulfuric, nitric) are secondary pollutants – ozone itself is a secondary pollutant of NOx photochemistry **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • The converse of any point, ie the same comparison written the other way round. **Do not credit** • carbon dioxide as the main cause of acid rain - the gases responsible are sulfur dioxide and oxides of nitrogen	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 237-238; AQA spec 3.4.1.	

Do not write outside the box

Question	Marking guidance	Marks
08.1	**Calculation** A correct final answer scores full marks on its own. If the answer is wrong, each correct step of working below earns its mark. **1** $3.6 \times 10^{11} / 1.2 \times 10^{9}$ **1** = 300 years **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (years) is already printed on the answer line. • **Residence time** → the time a molecule stays in the atmosphere / atmospheric lifetime • **aquifer** → a permeable rock that stores groundwater / underground water store **Do not credit** • A bare wrong answer with no working. • a dam described as having no environmental impacts - flooded land, blocked fish migration, trapped sediment and methane from drowned vegetation	2
08.2	**1 mark each** 1 mark for each of the 3 points below. **1** It must be porous, with spaces between the particles, so that it can store a large volume of water **1** It must be permeable, so that water can flow through it to a well or spring fast enough to be abstracted **1** An impermeable layer beneath it stops the water draining away, and the rock must be exposed at the surface somewhere, or overlain by permeable material, so that rainfall can recharge it **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **aquifer** → a permeable rock that stores groundwater / underground water store • **permeable** → how easily water passes through / lets water through • **impermeable** → does not let water through	3
08.3	**1 mark each** 1 mark for each of the 2 points below. **1** Abstraction of 4.8×10^{9} m^3 per year is four times the recharge of 1.2×10^{9} m^3 per year, so water is being removed faster than it is replaced **1** The stored water is being mined rather than used sustainably, so the saturated zone shrinks and its upper surface, the water table, falls each year **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **Abstraction** → water taken out of a river, lake or aquifer	2

Do not write outside the box

Question	Marking guidance	Marks
08.4	**Any 4 of 5** 1 mark each for any 4 of the 5 points below, in any order, maximum 4. **1** Wells and boreholes must be deepened and the water pumped from further down, so the energy cost of abstraction rises and shallow wells fail first **1** Rivers, springs and wetlands fed by the aquifer dry up, destroying habitats and the species that depend on them **1** Near the coast, sea water is drawn into the aquifer (saltwater intrusion), making the water too saline for drinking or irrigation **1** The rock compacts as the water that supported it is removed, causing land subsidence that damages buildings and drainage, and permanently reduces the storage capacity **1** Irrigated agriculture that depends on the aquifer fails, with loss of food production and livelihoods **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **aquifer** → a permeable rock that stores groundwater / underground water store • **abstraction** → water taken out of a river, lake or aquifer **Do not credit** • Points after the first four - the question asks for four, so a list of extras cannot pick up the marks.	4
08.5	**Any 4 of 5** 1 mark each for any 4 of the 5 points below, in any order, maximum 4. **1** Fossil water fell as rain thousands of years ago under a wetter climate – under the present climate recharge is negligible, so any abstraction depletes a finite store and the resource is effectively non-renewable **1** Changing to crops with a lower water demand, or to drip / trickle irrigation applied at night, which delivers water to the roots and loses far less to evaporation **1** Metering and pricing water, or setting abstraction licences and quotas, so that users have a reason to waste less **1** Recycling treated waste water for irrigation or industry, and harvesting rainwater, so less is taken from the aquifer **1** Lining canals and repairing leaking pipes so that less of the water abstracted is lost before it is used **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **aquifer** → a permeable rock that stores groundwater / underground water store • **abstraction** → water taken out of a river, lake or aquifer • **Recycling** → reprocessing a material to make something new **Do not credit** • 'renewables cause no pollution at all' - manufacture, construction and the land they occupy all have impacts • 'recycling uses no energy' - collection, sorting and reprocessing all use energy, usually less than making the material from new	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 115-121; AQA spec 3.2.2.1-3.2.2.2.	

Do not write outside the box

Question	Marking guidance	Marks
09.1	**Calculation** 1 mark for the correct answer; working does not have to be shown. **1** (120 + 95) - (2 + 45) = 168 t ha^{-1} **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (t ha^{-1}) is already printed on the answer line. **Do not credit** • A bare wrong answer with no working.	1
09.2	**1 mark each** 1 mark for each of the 2 points below. **1** Annual ploughing aerates the soil so aerobic decomposers break down dead organic matter faster, releasing CO2 **1** Harvest removes biomass so less dead organic matter is returned – pasture roots and litter constantly add organic matter **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
09.3	**Any 2 of 4** 1 mark each for any 2 of the 4 points below, in any order, maximum 2. **1** Low / zero tillage **1** Adding manure, compost or mulch **1** Growing cover crops / leaving crop residues **1** Converting to long-term grass or tree crops in rotation **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 148-151; AQA spec 3.2.4.2.	

Do not write outside the box

Question	Marking guidance	Marks
10.1	**Levels of response** Decide the level from the descriptors first, then use the indicative content to place the mark within that level. The points below are not a checklist. **Level 3 (7-9 marks)**: Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. **Level 2 (4-6 marks)**: Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. **Level 1 (1-3 marks)**: Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. **0 marks**: No relevant content. **Indicative content - a guide to placing the mark, not a checklist** • **Uses / benefits:** nuclear electricity (low carbon, high energy density), medicine (X-rays, CT, radiotherapy, tracers, sterilising equipment), industry (thickness gauges, polymer cross-linking, testing turbines, oil exploration), agriculture (food irradiation, sterile male technique, mutation breeding), research tracers, smoke detectors, ship / submarine propulsion • **Risks:** free radical production, DNA damage, mutation, cancer – acute and chronic, somatic and gonadic effects – contamination vs exposure – activation products – long-lived wastes – accidents (Chernobyl, Fukushima) – weapons fallout • **Factors affecting risk:** type of radiation (alpha, beta, gamma, neutrons - penetration, RBE / weighting factors), half-life, dose (Gy, Sv), route of exposure • Risk:benefit analysis and why it is difficult (latency, alternative causes, poor exposure data, risks and benefits to different people, offsets such as avoided surgery) • **Principles:** ALARA, BATNEEC – closed sources, remote handling, absorbers / shielding, distance (inverse square law), time, protective clothing, decontamination, potassium iodide • **Monitoring:** dosemeters, film badges, air and contamination monitors – environmental monitoring via CPA and CGM – waste categories and disposal (vitrification, encapsulation, Drigg) • **Evaluation / judgement:** benefits generally outweigh risks when controls are applied – public perception vs actual risk (background radon 50% of dose vs <1% from nuclear industry)	9

Do not write outside the box

Question	Marking guidance	Marks
	Also accept • Any other correct, relevant point of the same kind - wording never has to match these notes. • Any accurate, relevant material that is not in the indicative content. • **ionising radiation** → radiation with enough energy to remove electrons and form ions • **half-life** → the time for the activity, or the amount, to fall by half • **vitrification** → sealing waste into glass **Do not credit** • 'nuclear power releases no carbon dioxide' - mining, fuel processing, construction and decommissioning all do • 'after two half-lives nothing is left' - each half-life halves what remains, so some activity always continues	
	Model answer - Level 3 (8 marks)	
	Ionising radiation is used because it does things nothing else can: nuclear fission generates low-carbon electricity from fuel of extraordinary energy density; medicine relies on X-rays and CT scans, radiotherapy, tracers and gamma sterilisation of equipment; industry uses thickness gauges, cross-links polymers, tests turbine blades and explores for oil; and agriculture uses irradiation to preserve food and sterilise insect pests. The risks arise because radiation produces free radicals that damage DNA, causing mutation and cancer, with acute and chronic, somatic and gonadic effects; contamination is worse than exposure because the source stays with the person; and activation products, long-lived wastes, accidents at Chernobyl and Fukushima and weapons fallout add environmental risks.	
	Risk depends on the type of radiation - alpha is barely penetrating but has a weighting factor of 20 - on half-life, on dose in grays and sieverts and on the route of exposure. Weighing risks against benefits is difficult because cancers appear years later with many other possible causes, exposure data are poor, the people who benefit are often not those at risk, and benefits such as avoided surgery are hard to quantify.	
	Control follows ALARA and BATNEEC: closed sources, remote handling, shielding, distance under the inverse-square law, limiting time, protective clothing, decontamination and potassium iodide against iodine-131. Workers wear dosemeters and film badges and pass contamination monitors; discharges are managed by Critical Pathway Analysis and Critical Group Monitoring; wastes are vitrified, encapsulated or stored at licensed sites. With these controls the benefits generally outweigh the risks - natural radon supplies about half the average dose and the nuclear industry under one per cent - although public perception of risk remains far higher than the measured risk.	
	Weaker answer - Level 2 (5 marks)	
	Ionising radiation is used in nuclear power stations to make electricity, in hospitals for X-rays and treating cancer, and in industry. The risks are that radiation damages DNA and causes cancer, and accidents like Chernobyl release radiation into the environment. Nuclear waste stays dangerous for a long time. Workers are protected by shielding, keeping their distance and wearing film badges to measure their dose. The public is protected by monitoring around nuclear sites. Overall the benefits are worth the risks if it is controlled properly.	
	What makes the difference	

Do not write outside the box

Question	Marking guidance	Marks
	• Level 2 covers uses, risks, controls and a judgement, but each briefly; Level 3 adds the factors that determine risk (type, half-life, dose units, route) and why risk-benefit analysis is difficult.	
	• Level 3 names the principles (ALARA, BATNEEC, inverse square law, CPA / CGM) rather than describing practice loosely.	
	• The judgement in Level 3 is supported with evidence (radon vs nuclear industry doses); Level 2's is an opinion.	
10.2	1 mark for the point below. **1** Consumers of the electricity may live in a different country / region from the uranium miners or the communities near waste stores who bear the risks **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • 'nuclear power releases no carbon dioxide' - mining, fuel processing, construction and decommissioning all do	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 295-304; AQA spec 3.4.3.2.14.	

Do not write outside the box

Question	Marking guidance	Marks
11.1	**Discuss the extent to which the study of ozone depletion illustrates how scientific investigation can lead to the successful solution of an environmental problem.** **Levels of response** **Level 5 (21-25 marks)**: Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. **Level 4 (16-20 marks)**: Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. **Level 3 (11-15 marks)**: Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. **Level 2 (6-10 marks)**: Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. **Level 1 (1-5 marks)**: Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. **0 marks**: No relevant content. **Indicative content - students are not expected to cover all of these** • **Stages of investigation:** identifying the issue (Halley Station UV data, Dobson unit measurements), hypothesis formation (Rowland-Molina 1974: persistence, UV dissociation, chlorine reactions), data collection (ground, satellite, balloon and aircraft air sampling detecting ClO), analysis of spatial, temporal and altitude variability, proposal of solutions, enactment • **Why Antarctica:** low stratospheric temperatures, ice crystals, stratospheric clouds, polar vortex, spring sunlight • **Effects of UVB:** skin cancer, cataracts, DNA damage, plant and marine damage - motivating action • **Solutions:** Vienna Convention, Montreal Protocol - phase-out and bans, alternative processes (pump sprays), alternative materials (HCFCs, HFCs, hydrocarbons, HFAs), collection and incineration of CFCs, fund for developing countries • **Evaluation of success:** ozone hole area / depth stabilising, CFC levels falling	25
	– limitations - persistence, remaining ODSs, HFCs as greenhouse gases, decades to full recovery • Comparison with other atmospheric problems (climate change, acid rain) - why ozone was easier – role of peer review, international cooperation, availability of substitutes • Conclusion on the extent to which it is a model for other issues *Credit any relevant, accurate material, including points that are not listed here. Reward a clear line of argument and a supported conclusion.* **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 108-113; AQA spec 3.2.1.3.	

Do not write outside the box

Question	Marking guidance	Marks
11.2	**Discuss the extent to which energy conservation technologies could reduce the need for new energy supplies in a developed country.** **Levels of response** **Level 5 (21-25 marks)**: Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. **Level 4 (16-20 marks)**: Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. **Level 3 (11-15 marks)**: Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. **Level 2 (6-10 marks)**: Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. **Level 1 (1-5 marks)**: Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. **0 marks**: No relevant content. **Indicative content - students are not expected to cover all of these** • **Scale of the problem:** how energy is used (transport, buildings, industry, water) – links between affluence and per capita use • **Transport:** bulk / integrated transport, ATM, vehicle design (aerodynamics, mass, tyres, KERS, combustion efficiency), low embodied energy, design for end of life, driver behaviour, electric vehicles • **Buildings:** orientation and passive solar, SA:V, thermal mass, low embodied energy materials, earth sheltering, glazing (low-e, inert gas, vacuum), insulation optimum, heat exchangers, automatic ventilation, occupancy sensors, thermostats, low-energy appliances, behaviour • **Industry:** heat recovery and exchangers, insulation and high-volume storage, CHP, integrated manufacture, recycling and mass reduction, high-voltage grid, peak shaving, ICT management • **Limits:** rebound effect, cost and payback, existing building stock, embodied energy of replacements, behaviour • Evaluation with quantitative examples – comparison with the cost of new supply	25
	– conclusion that conservation is the cheapest first step but cannot alone meet demand growth *Credit any relevant, accurate material, including points that are not listed here. Reward a clear line of argument and a supported conclusion.* **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 170-174, 225-235; AQA spec 3.3.4.2, 3.3.1.	

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